

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Anchal Kumari(192312255) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:

Question 1 — Case Study 1: Smart Medical Diagnosis System

Q1(a) Propositional Logic Representation

Symbols used:

- F = Patient has Fever
- C = Patient has Cough
- H = Patient has Rash
- B = Patient has Breathlessness
- L = Possible Flu
- M = Possible Measles
- P = Possible Pneumonia

Rules:

- R1: $F \wedge C \rightarrow L$
- R2: $F \wedge H \rightarrow M$
- R3: $C \wedge B \rightarrow P$

Facts (Patient A): F = True, C = True, B = True, H = False / Unknown

Q1(b) Forward Chaining

Step	Rule Checked	Condition	Result
1	R1: $F \wedge C \rightarrow L$	$F = T, C = T \rightarrow$ satisfied	$\text{Fire} \rightarrow L \text{ (Flu)} = \text{True}$
2	R2: $F \wedge H \rightarrow M$	H unknown/False $\rightarrow$ not satisfied	Not fired
3	R3: $C \wedge B \rightarrow P$	$C = T, B = T \rightarrow$ satisfied	$\text{Fire} \rightarrow P \text{ (Pneumonia)} = \text{True}$

Derived facts: L (Possible Flu), P (Possible Pneumonia)

Q1(c) Backward Chaining — Goal: P (Pneumonia)

Goal: P +-- needs: C AND B (Rule III) +-- C -> check KB -> Fact: C = True [OK] +-- B -> check KB -> Fact: B = True [OK] Both subgoals satisfied $\Rightarrow P = \text{True}$ (GOAL PROVED)
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Question 2 — Case Study 2: Autonomous Traffic Management Agent

Q2(a) Unification with Rule 1

Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

- Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance}) \rightarrow \theta_1 = \{x/\text{Ambulance}\}$
- Unify $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$ under $\theta_1 \rightarrow$ consistent
- MGU: $\theta = \{x/\text{Ambulance}\}$
- Apply to conclusion $\rightarrow \text{ClearPath}(\text{Ambulance})$ derived

Q2(b) Resolution Proof — Prove Proceed(CarA)

CNF of the rules:

- 1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
- 2. $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$
- 3. $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses): $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal: $\neg \text{Proceed}(\text{CarA})$

Resolution steps:

1) (Clause 1)+$\text{Vehicle}(\text{Ambulance}) \{x/\text{Ambulance}\}$ $\rightarrow \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$ 2) + $\text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$ 3) (Clause 2){$y/\text{Ambulance}$} + $\text{ClearPath}(\text{Ambulance}) \rightarrow \text{GreenSignal}(\text{Ambulance})$ 4) (Clause 3){$x/\text{CarA}, y/\text{Ambulance}$} + $\text{Vehicle}(\text{CarA})$ $\rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

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5) + Behind(CarA,Ambulance) -> ¬GreenSignal(Ambulance) ∨ Proceed(CarA)
6) + GreenSignal(Ambulance) -> Proceed(CarA)
7) + ¬Proceed(CarA) (negated goal) -> [] EMPTY CLAUSE

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Conclusion: Proceed(CarA) is proved TRUE.

Q2(c) Sufficiency of the KB for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient. Gaps identified:

- No predicate for priority/ordering between two competing ambulances at the same junction.
- No rule to resolve conflict when two ClearPath requests intersect (e.g., perpendicular roads).
- Needed additions: Priority(x), Conflict(x,y), and a rule such as: $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{Conflict}(x,y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$.
- Also needs timestamp/arrival-order facts to break ties when priorities are equal.

Question 3 — Case Study 3: Agricultural Expert Reasoning System

Q3(a) CNF Conversion

Symbols: D = SoilDry, I = IrrigationNeeded, W = CropWheat, A = ApplyDripMethod, R = CropAtRisk

- P1: $D \rightarrow I \Rightarrow \neg D \vee I$
- P2: $I \wedge W \rightarrow A \Rightarrow \neg(I \wedge W) \vee A \Rightarrow \neg I \vee \neg W \vee A$
- P3: $\neg A \wedge W \rightarrow R \Rightarrow \neg(\neg A \wedge W) \vee R \Rightarrow A \vee \neg W \vee R$
- Facts: D, W (already unit clauses)

Q3(b) Resolution Refutation — Prove ApplyDripMethod (A)

Clauses: (1) $\neg D \vee I$ (2) $\neg I \vee \neg W \vee A$ (3) $A \vee \neg W \vee R$ (4) D (5) W (6) $\neg A$ (negated goal)

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D (4) + ¬D ∨ I (1)          -> I
I + ¬I ∨ ¬W ∨ A (2)         -> ¬W ∨ A
¬W ∨ A + W (5)              -> A
A + ¬A (6, negated goal)    -> [] (empty clause)

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Conclusion: ApplyDripMethod (A) is proved TRUE.

Q3(c) If Fact 'SoilDry' (D) is Removed

- P1 ($D \rightarrow I$) can no longer fire, so I cannot be derived.
- P2 ($I \wedge W \rightarrow A$) cannot fire without I, so A cannot be proven true.
- Under strict classical logic: A is undetermined (neither proved true nor false), so P3's antecedent $\neg A \wedge W$ cannot be confirmed either $\rightarrow R$ (CropAtRisk) remains undetermined, not proven.
- Under the Closed World Assumption (common in expert systems — 'if it cannot be proven true, treat as false'): failure to derive $A \Rightarrow$ assume $\neg A$. Then $\neg A \wedge W(\text{True}) \rightarrow R$ fires, so $\text{CropAtRisk} = \text{True}$ still follows — but only via this non-monotonic reasoning pattern, not classical entailment.

Question 4 — Case Study 4: Wumpus World Knowledge Agent

Q4(a) Belief State via Modus Ponens

Percepts: Stench[1,2] = True, Breeze[1,1] = True, Glitter[2,2] = True

Rule	Premise	Modus Ponens Applied	Derived Fact
1	Stench[1,2] = T	Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]	WumpusAdjacent[1,2] = True
2	Breeze[1,1] = T	Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]	PitAdjacent[1,1] = True
3	Glitter[2,2] = T	Glitter[2,2] $\rightarrow$ Gold[2,2]	Gold[2,2] = True

Derived conclusions: WumpusAdjacent[1,2], PitAdjacent[1,1], Gold[2,2]

Q4(b) Forward Chaining — Locating the Wumpus and the Pit

- Neighbours of [1,2] are {[1,1], [1,3], [2,2]}. The agent is alive at [1,1] with no stench recorded there, so Wumpus $\neq$ [1,1] $\rightarrow$ Wumpus $\in$ {[1,3], [2,2]}.
- Neighbours of [1,1] (corner cell) are {[1,2], [2,1]}. Since Breeze[1,1] is true $\rightarrow$ Pit $\in$ {[1,2], [2,1]}.
- Applying Rule 4 ($\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]): cell [1,1] itself is Safe (the agent survived entering it); no other cell can yet be certified Safe since the candidate sets above remain ambiguous.

Q4(c) Is the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] Safe?

- [1,2]: Cannot rule out a Pit — it is one of the two Pit candidates identified in Q4(b). Rule 4 cannot certify it Safe $\rightarrow$ risk of Pit.
- [2,2]: It is one of the two Wumpus candidates identified in Q4(b) $\rightarrow$ risk of Wumpus, despite containing the Gold.

Conclusion: The path is NOT logically proven safe. Neither [1,2] nor [2,2] satisfies $\neg$ Wumpus $\wedge$ $\neg$ Pit, so Rule 4 cannot assert Safe for either cell. The agent should first visit [2,1] or [1,3] to gather additional Breeze/Stench percepts and narrow down the Pit/Wumpus locations via further forward chaining before committing to this path.